Lecture 1

Recap and Basics

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A refresher

- A celebrated example
- Strategic form games: formalism
- Nash Equilibria
- Applications
- Extensive form games: formalism
- Subgame perfect equilibrium
- Applications

I know you've seen this

India and Pakistan can either

- Develop a nuclear weapon (D)
- Not develop a nuclear weapon ($\neg D$)

Of course, their well-being depends on their own action, as well as their rival's ($\Rightarrow$ strategic interaction)

Both better off when no one develops weapons compared to both developing weapons (same balance of power, <$)

But neither wants to be *the only one* without nukes

What is going to happen?

I know you've seen this

		Pakistan	
		$\neg D$	D
India	$\neg D$	$2, 2$	$0, 3$
	D	$3, 0$	$1, 1$

- What do these numbers mean?
- Is it a descriptively accurate representation of the IND-PAK rivalry?

Informal Analysis

There are no incentives to uphold an agreement in which both countries choose $\neg D$:

- If IND believes PAK will violate the agreement, it has an incentive to violate the agreement
- Even if IND believes that PAK will uphold the agreement, it has an incentive to violate the agreement

The only scenario we can reasonably expect is that both countries choose D

Even though they would both prefer both choosing $\neg D$

The Prisoner's Dilemma is **a way** to formalize this strategic interaction

Strategic form (or simultaneous move) games

- A set of strategic[1] players $N := \{1, ..., n\}$
- For each player $i \in N$, a set of actions A_i
  - $a_i \in A_i$ is the generic element of A_i
  - $a = (a_1, ..., a_n) \in A_1 \times ... \times A_n$ is a generic action profile
- For each player, a payoff function $\pi_i : A_1 \times ... \times A_n \to \mathbb{R}$

Strategic interaction: for at least some player, the function $\pi_i(a)$ is non-constant in some other player's j action

A pure strategy for i is a choice of an action

An action profile chosen by players is called an **outcome**

1. Non-strategic players' behavior is assumed

In the previous game

- $N := \{IND, PAK\}$
- $A_{IND} = A_{PAK} = \{\neg D, D\}$
- Payoff function

  $\pi_{IND}(D, \neg D) = \pi_{PAK}(\neg D, D)$ = 3
  $\pi_{IND}(\neg D, \neg D) = \pi_{PAK}(\neg D, \neg D)$ = 2
  $\pi_{IND}(D, D) = \pi_{PAK}(D, D)$ = 1
  $\pi_{IND}(\neg D, D) = \pi_{PAK}(D, \neg D)$ = 0

What is going to happen? More tools

A (mixed) **strategy** for player i is a probability distribution over her set of actions: $\sigma_i : 2^{A_i} \to [0,1]$

- $\sigma_{IND}(D)$ is probability assigned by σ_{IND} to D

A strategy profile $\sigma = \{\sigma_1, ..., \sigma_n\}$ is a collection of strategies for each player (NB: no correlation)

Player i's expected payoff from σ is given by

$$u_i(\sigma) = \sum_{(a_1,...,a_n)\in A_1\times...\times A_n} \left(\prod_{j\in N} \sigma_j(a_j)\right) \pi_i(a_1, ..., a_n)$$

Notation $\sigma \equiv (\sigma_i, \sigma_{-i})$, σ_{-i}

is strategy of everyone *but i*

Exercise

In the India-Pakistan version of the prisoner's dilemma, consider the following mixed strategy profile

$$\sigma = \left[\begin{array}{ll} \sigma_{IND}(D) = 0.3, & \sigma_{IND}(\neg D) = 0.7 \\ \sigma_{PAK}(D) = 0.7, & \sigma_{PAK}(\neg D) = 0.3 \end{array} \right]$$

For each player, compute $u_{IND}(\sigma)$ and $u_{PAK}(\sigma)$, the expected utility associated with this profile

Nash equilibrium

σ_i is a **best response** to σ_{-i} if

$$u_i(\sigma_i, \sigma_{-i}) \geq u_i(\sigma_i', \sigma_{-i}) \quad \forall \quad \sigma_i' \neq \sigma_i$$

Definition

Strategy profile $\sigma = (\sigma_1, .., \sigma_n)$ is a **Nash equilibrium** if for each i, σ_i is a best response to σ_{-i}

A Nash equilibrium σ is immune to **profitable deviations**: no player can improve her payoff by choosing a $\sigma_i'' \neq \sigma_i$ such that

$$u_i(\sigma_i'', \sigma_{-i}) > u_i(\sigma)$$

Intuition

We posit that players “respond to incentives:” *given how they expect their opponents to behave*, they choose what’s best for them

Plausibility of this rationality assumption depends on how actions and payoffs are specified

Interpretation of Nash Equilibrium

No unilateral incentive to change behavior

Consistent with a notion of stability:

- players can predict a NE profile, predict their opponents predict it, and so on
- If players predict a non-NE profile, someone is making a mistake and should change behavior
- For the analyst to predict a non-NE profile, analyst must know more about the game than participants

Social learning

So why is this called Nash?

A strategic form game is **finite** if all players have finite A_i

Theorem

Every finite strategic form game has a Nash equilibrium

Theorem

Every strategic form game whose

- action sets are nonempty closed and bounded subset of an Euclidean space
- payoff functions are continuous

has a Nash equilibrium

Finding Nash equilibria

Iterated deletion of strictly dominated actions (IDSA)

Step 1

Is there a player i, an action $a'_i \in A_i$, and a strategy σ'_i such that $\forall a_{-i} \in \prod_{j \in N \setminus \{i\}} A_j \; u_i(a'_i, a_{-i}) < u_i(\sigma'_i, a_{-i})$?

Step 2a

If so, $\sigma_i(a'_i) = 0$ (eliminate a'_i from A_i), back to Step 1.

Step 2b

If not, procedure ends

Step 3

Look for Nash equilibria in this game with restricted sets of actions

Every strategy σ_i in a Nash equilibrium σ puts positive probability only on actions that survive IDSA

Finding Mixed Strategy Nash Equilibria

Using IDSA to find mixed strategy Nash equilibria

The augmented matching pennies game:

		Pl. 2			
		H	T	L	R
Pl. 1	H	$1, -1$	$-1, 1$	$0, -2$	$0, -3$
	T	$-1, 1$	$1, -1$	$0, -2$	$0, -3$
	D	$-2, 0$	$-2, 0$	$-2, 1$	$2, 0$

Finding Mixed Strategy Nash Equilibria

Using IDSA to find mixed strategy Nash equilibria

Augmented matching pennies

Pl. 2

Pl. 1		H	T	L	$\cancel{R}$
	H	$1,-1$	$-1,1$	$0,-2$	$\cancel{0,-3}$
	T	$-1,1$	$1,-1$	$0,-2$	$\cancel{0,-3}$
	D	$-2,0$	$-2,0$	$-2,1$	$\cancel{2,0}$

Finding Mixed Strategy Nash Equilibria

Using IDSA to find mixed strategy Nash equilibria

Augmented matching pennies

Pl. 2

Pl. 1		H	T	L
	H	$1, -1$	$-1, 1$	$0, -2$
	T	$-1, 1$	$1, -1$	$0, -2$
	$\cancel{D}$	$\cancel{-2, 0}$	$\cancel{-2, 0}$	$\cancel{-2, 1}$

Finding Mixed Strategy Nash Equilibria

Using IDSA to find mixed strategy Nash equilibria

Augmented matching pennies

		Pl. 2		
		H	T	$\cancel{L}$
Pl. 1	H	$1, -1$	$-1, 1$	$\cancel{0, -2}$
	T	$-1, 1$	$1, -1$	$\cancel{0, -2}$

Finding Mixed Strategy Nash Equilibria

		Pl. 2	
		H	T
Pl. 1	H	$1, -1$	$-1, 1$
	T	$-1, 1$	$1, -1$

If a strategy is a NE, for each player it must place positive probability on H and T only

How do players mix?

		Pl. 2	
		H	T
Pl. 1	H	$1, -1$	$-1, 1$
	T	$-1, 1$	$1, -1$

Suppose Pl. 2 plays H with probability $\sigma_2(H) \in (0, 1)$

Then Pl. 1 must also mix (why?) Then $\sigma_2(H)$ must make Pl. 1 **indifferent** between playing H or T:

$$u_1(H, \sigma_2) = 2\sigma_2(H) - 1$$
$$u_1(T, \sigma_2) = 1 - 2\sigma_2(H)$$

$\Rightarrow \sigma_2(H) = 1/2$. By the same reasoning, $\sigma_1(H) = 1/2$

Exercise

Apply IDSA to this game

		Pl. 2	
		L	R
Pl. 1	U	$2, 1$	$-1, 2$
	M	$0, 3$	$0, 1$
	D	$-1, -2$	$2, -1$

Exercise

Find mixed strategy equilibria of this game

Pl. 2

Pl. 1		R	P	S
	R	$0, 0$	$-1, 1$	$1, -1$
	P	$1, -1$	$0, 0$	$-1, 1$
	S	$-1, 1$	$1, -1$	$0, 0$

Multiplicity

Of course, with multiple Nash equilibria, no sharper prediction than “each equilibrium can happen”

Pl. 2

Pl. 1		R	P
	R	$2, 2$	$0, 0$
	P	$0, 0$	$1, 1$

Which equilibrium is more plausible?

How about this one?

Pl. 2

Pl. 1		R	P
	R	$2, 2$	$-100, 0$
	P	$0, -100$	$1, 1$

The (modified) Hotelling-Downs game

Two candidates, L and R, and a continuum of voters

Each candidate $j \in \{L, R\}$ chooses a "platform" $p_j \in [0, 1]$

Candidates' payoff is 1 when elected, zero otherwise

Each voter i characterized by ideal policy $p_i \in [0, 1]$; cdf $F(\cdot)$ describes distribution of ideal policies

Each voter i can either vote for R or L, and we assume she votes for the candidate whose platform is closest to p_i

Candidate with most votes wins, otherwise winner determined by the flip of a coin

Where do candidates locate?

The (modified) Hotelling-Downs game

Let p^m be the median policy position (solution to $F(p) = 1/2$)

We argue that there exists [why isn't this obvious?] a unique equilibrium in which $p_R = p_L = p^m$

let $\pi_j(p_R, p_L) \in \{0, 1\}$ be candidate j's payoff given a profile (p_R, p_L)

let $\Pi_j(\sigma_R, \sigma_L) \in [0, 1]$ be candidate j's expected payoff from mixed strategy profile $(\sigma_R,, \sigma_L)$[2]

2. slightly abusing notation, $\sigma_j = p$ represents the strategy $\sigma_j(p) = 1$

The (modified) Hotelling-Downs game

Claim 1. *There exists a unique Nash equilibrium (NE):*

$$\sigma_R = \sigma_L = p^m$$

Proof

1. $\sigma_R = p^m$ has no profitable deviation when $\sigma_L = p^m$, since $\Pi_R(\sigma_R, p^m) < 1/2 \quad \forall \sigma_R \neq p^m$

2. If $\sigma_R(P') > 0$, with $p^m \notin P'$, then (σ_R, σ_L) is not a NE

   - if $\Pi_R(\sigma_R, \sigma_L) \geq 1/2$
   - then it must be that $\sigma_L \neq p^m$
   - and p^m is a profitable deviation for L, since $\Pi_L(\sigma_R, p^m) > 1/2$ and thus $\Pi_R(\sigma_R, p^m) < 1/2$

3. Use the same logic of step 2. to rule out $\sigma_L(P') > 0$, with $p^m \notin P'$

Proof

1. $\sigma_R = p^m$ has no profitable deviation when $\sigma_L = p^m$, since $\Pi_R(\sigma_R, p^m) < 1/2 \quad \forall \sigma_R \neq p^m$

2. If $\sigma_R(P') > 0$, with $p^m \notin P'$, then (σ_R, σ_L) is not a NE

- if $\Pi_R(\sigma_R, \sigma_L) < 1/2$
- then it must be that $\sigma_R = p^m$ is a profitable deviation for R
- why? for all σ_L, $\Pi_R(p^m, \sigma_L) \geq 1/2$

3. Use the same logic of step 2. to rule out $\sigma_L(P') > 0$, with $p^m \notin P'$

Exercise (Calvert-Wittman)

Candidates are policy-motivated: they care only about the final policy p

Candidate L's payoff equals to $-p$, R's payoff is $-(1-p)$

As before, each candidate is committed to implementing the policy announced: p is either the realization of σ_R or the realization of σ_L

Find the pure strategy Nash equilibrium of this game

Extensive form (multi-stage) games

A peasant must decide whether to plant a unit of corn (invest) or to eat the corn (consume)

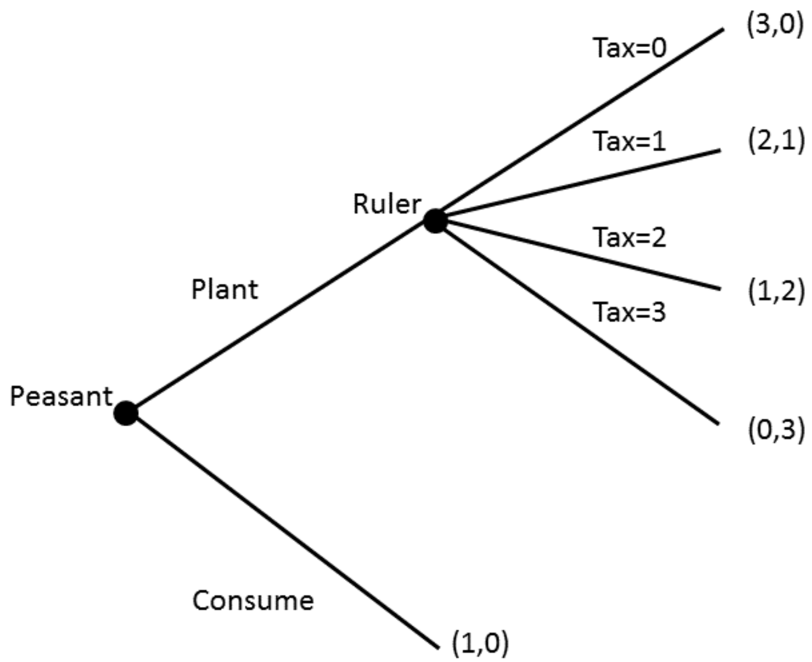
If he plants the corn, he gets 3 units. A ruler, however, can set a tax $\in \{0, 1, 2, 3\}$ on the corn production

The peasant prefers to consume more corn to less corn, and the ruler prefers to expropriate more to less corn

This interaction can be modeled as a extensive form game of complete information

Here, the order of play is substantively important: first, the peasant makes investment/consumption decision, then ruler decides expropriation

A formal representation



What happens?

Time consistency problem: due to the timing of the game, a commitment by Ruler not to expropriate lacks credibility

Unique pure strategy Nash equilibrium

Relationship between action and strategy is more complicated: analysis requires additional machinery

Specifically, we need to define **histories, outcomes** and **strategies**

Precise general notation for sequential games is quite cumbersome, we'll keep it simpler

Histories

A **history** specifies all of the actions that were taken by each player that moved in previous stages of the game

Stages that comes after different histories are considered to be different stages

If no player moves after a certain history, we say that the history is terminal, and refer to it as an **outcome**

If at each non-terminal history one (and only one) player chooses an action, the game has **perfect information**

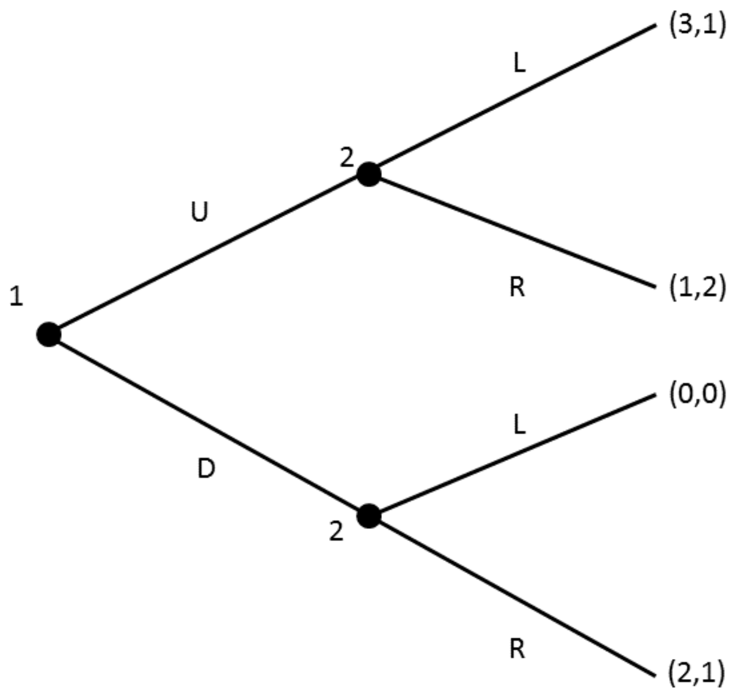
(Perfect information) extensive form game

- A set of strategic[3] players $N := \{1, ..., n\}$
-
- A set of histories $\mathcal{H} = \{h_1, ..., h_H\}$ (what happened up to that point)—we denote by $\mathcal{O} \subset \mathcal{H}$ the set of terminal histories, or outcomes
- For each player i, action space after each history $A_i(h)$
- For each player, a payoff function $\pi_i : \mathcal{O} \to \mathbb{R}$

A **pure strategy** is, for each player, a complete plan of action (what I do at each stage in which I choose)

3. There may be non-strategic players whose behavior is assumed

Another example



Another example - continued

Here we have

- $\mathcal{H} = \{\circ, U, D, UL, UR, DL, DR\}$
- $A_1(\circ) = \{U, D\}$
- $A_2(U) = \{L, R\}$
- $A_2(D) = \{L, R\}$
- Pure Strategies:
    - $S_1 = A_1(\circ)$
    - $S_2 = \{L(U)L(D), L(U)R(D), R(U)L(D), R(U)R(D)\}$

Another example

Pl. 2

		$L(U)L(D)$	$L(U)R(D)$	$R(U)L(D)$	$R(U)R(D)$
Pl. 1	U	$3, 1$	$3, 1$	$1, 2$	$1, 2$
	D	$0, 0$	$2, 1$	$0, 0$	$2, 1$

$(U, R(U), L(D))$ is not a “plausible” Nash

Why? Pl. 2’s “threat” of playing L after D is not credible

From Nash to Subgame Perfect equilibrium

This notion of strategy implicitly endows players with an unrealistic amount of commitment

Nash equilibrium becomes usually unable to produce sharp predictions

A **subgame** is a game that begins after each non-terminal history

A **subgame perfect equilibrium (SPE)** is a strategy profile that induces a Nash equilibrium in every subgame

Properties of SPE

Idea is older than Nash Equilibrium itself ; modern formal definition: ;

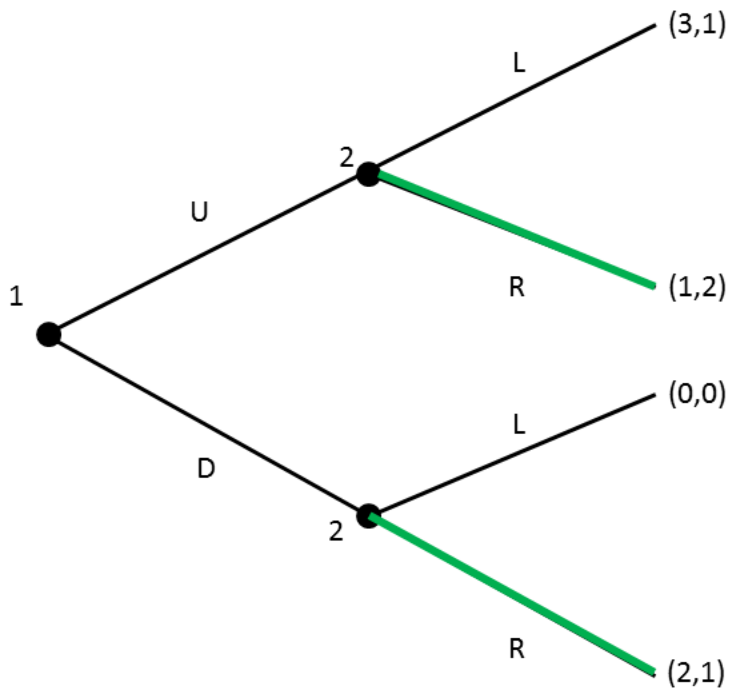
An extensive form game (EFG) is **finite** if after every history, each player that moves has a finite set of actions

Theorem

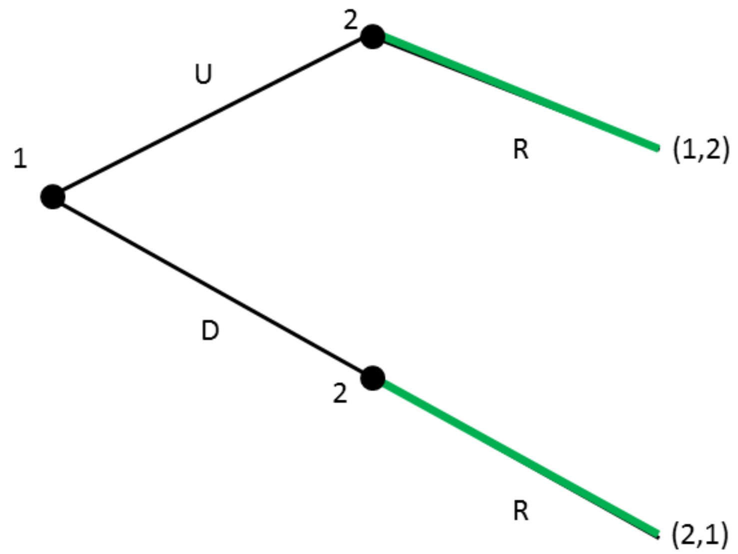
1. If a finite EFG with perfect information has a finite number of stages, then it has a pure strategy SPE.

2. If, in addition, for each player each outcome has a different payoff, then the SPE is unique

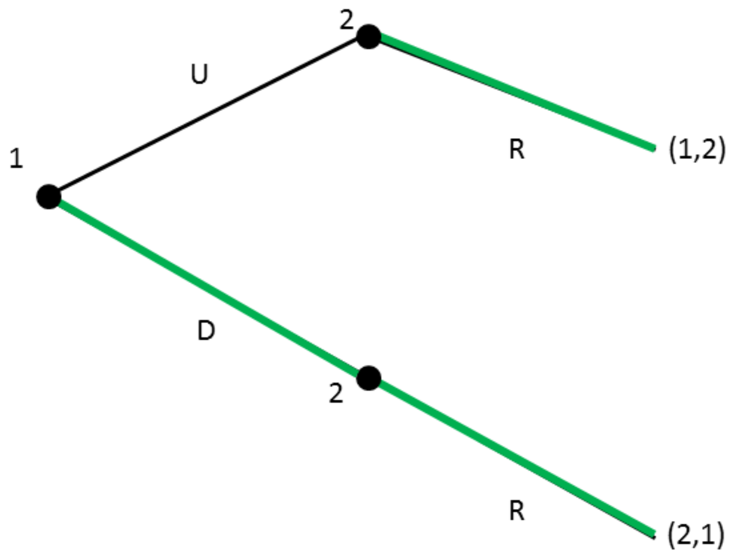
Use of backward induction to find SPE



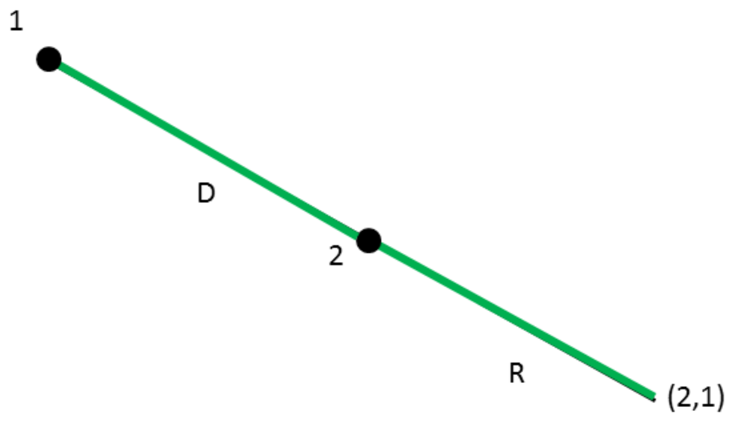
Use of backward induction to find SPE



Use of backward induction to find SPE



Use of backward induction to find SPE



Technical note

What is a mixed strategy in an extensive form game?

If a player is indifferent at a particular stage and mixes, not a mixed strategy in the conventional sense

Yet, studying conventional mixed strategies or these **behavioral strategies** (a distribution associated to each stage) leads to the same conclusion

Mixed strategies in the previous game

A behavioral strategy: Player 2 chooses two numbers

$$\sigma(U) = Pr(a_1 = U)$$
$$\sigma(R \mid U) = Pr(a_2 = R \mid a_1 = U)$$
$$\sigma(R \mid D) = Pr(a_2 = R \mid a_1 = D)$$

A mixed strategy: Player 2 chooses three numbers

$$\sigma(U) = Pr(s_1 = U)$$
$$\sigma(L(U)L(D)) = Pr(s_2 = L(U)L(D))$$
$$\sigma(L(U)R(D)) = Pr(s_2 = L(U)R(D))$$
$$\sigma(R(U)L(D)) = Pr(s_2 = R(U)L(D))$$

An Example

In the previous game, the unique SPE is given by:

$$\sigma^* = [\sigma(U) = 0, \quad \sigma(R \mid U) = 1, \quad \sigma(R \mid D) = 1]$$

Exercise

In the previous game, change player 2's payoffs as follows:
$\pi_2(D, R) = 1$, $\pi_2(U, L) = \pi_2(U, R) = \pi_2(D, L) = 0$

Is σ^* still a SPE in this modified game?

To characterize the full set of SPE, consider three cases separately

1. 1 is indifferent ($\sigma(U) \in [0, 1]$)
2. 1 strictly prefers U ($\sigma(U) = 1$)
3. 1 strictly prefers D ($\sigma(U) = 0$)

For each case, ask (i) what must be true about $\sigma(R \mid U)$ and $\sigma(R \mid D)$ and (ii) is this individually rational (payoff-maximizing) for 2?

Exercise (the ultimatum game)

Two players, 1 and 2, who must decide how to share a unit of surplus; payoffs are linear in the obtained share

First, 1 makes an offer to player 2, then 2 decides whether or not to accept

Let $x \in [0, 1]$ be 1's offer, and $\{A, R\}$ be 2's action space

If 2 accepts, surplus is split, otherwise it is destroyed:

$$u_1(x, A) = 1 - x, \quad u_2(x, A) = x$$
$$u_1(x, R) = u_2(x, R) = 0$$

The game has a unique SPE. Why?

Why is this statement not obvious?

Exercise (the first mover)

Find the SPE of the matching pennies game in which player 1 moves first, and then player 2 moves

Find the SPE of the Cournot-Tullock Contest in which first Pl. 1 chooses effort $a1$, then Pl. 2 chooses effort $a2$

Find the SPE of the Hotelling-Downs model in which L moves first

Compare all of the SPEs above with the Nash equilibria of the strategic form versions of these games

Matching Pennies

		Pl. 2	
		H	T
Pl. 1	H	$1, -1$	$-1, 1$
	T	$-1, 1$	$1, -1$

The Cournot-Tullock contest

Two gangsters, $DonFalcone$ and $FishMooney$, must decide how much to invest in competing for a resource (city crime hegemony) of value $R > 0$

Each player $i \in \{DonFalcone, FishMooney\}$ can invest any amount $a_i \geq 0$ at cost c_i

The probability that i wins (denote by $-i$ his opponent) is[4]

$$p_i(a_i, a_{-i}) = \frac{\lambda_i a_i}{\lambda_i a_i + \lambda_{-i} a_{-i}}$$

Each player's expected payoff is

$$u_i(a_i, a_{-i}) = Rp_i(a_i, a_{-i}) - c_i a_i$$

4. Assume for simplicity that $p_i(0,0) = \frac{\lambda_i}{\lambda_i + \lambda_{-i}}$

References